

Notes: Favara & Imbs (AER, 2015)

Credit Supply and the Price of Housing

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1 Big picture

- **Question.** Does an expansion in mortgage credit supply raise house prices, or do house prices merely cause more credit through collateral and demand channels?
- **Identification strategy.** The paper uses post-1994 interstate branching deregulation as an instrument for mortgage credit supply, but does not simply assume deregulation is exogenous. Instead, it compares treated banks to carefully chosen placebo lenders and location-based control groups.
- **Credit-supply shock.** Branching deregulation lets banks diversify deposits geographically and lower funding costs. Commercial banks covered by the regulation expand mortgage credit; legally unaffected lenders do not.
- **Main mortgage-credit result.** Deregulation increases the number and volume of commercial-bank mortgage originations and raises loan-to-income ratios, while denial rates and loan sales do not move much.
- **Key placebo test.** Independent mortgage companies, thrifts, and credit unions do not respond in the same way. This weakens demand-based explanations.
- **Location test.** Out-of-state banks with branches in a deregulating state respond strongly; out-of-state banks without local branches and in-state banks do not.
- **House-price result.** Deregulation raises house prices, especially where housing supply is inelastic. In more elastic areas, deregulation increases the housing stock instead.
- **IV result.** Instrumented mortgage credit growth explains house-price growth: an expansion in credit supply causes higher house prices.
- **Magnitude.** Between 1994 and 2005, deregulation explains about one-half to two-thirds of the increase in commercial-bank mortgage loans and about one-third to one-half of the increase in house prices.
- **Mechanism evidence.** Deregulation reduces mortgage rates and deposit costs, increases deposit growth, and does not raise nonperforming loans.
- **Economic takeaway.** Credit supply matters for asset prices. House-price booms can be driven not only by demand or collateral feedback, but also by bank funding and regulatory shocks that expand mortgage supply.

2 Data and measurement

2.1 Mortgage-credit data

- Mortgage data come from HMDA and cover US urban counties from 1994 to 2005.
- The paper separates commercial banks from independent mortgage companies, thrifts, and credit unions.
- Key mortgage outcomes are:
 - number of applications received;
 - number of loans originated;
 - dollar volume of mortgage loans;
 - number of denials;
 - loan-to-income ratio;
 - number of loans sold or securitized.

Interpretation: Commercial banks are the treated lender group because interstate branching deregulation affects their ability to branch and fund loans. Independent mortgage companies and other nonbank lenders provide placebo samples.

2.2 Deregulation measure

- The policy variation is the Rice and Strahan index of interstate branching deregulation.
- The index ranges from 0 to 4.
- A higher value means fewer restrictions on interstate branching.
- The timing and intensity of deregulation vary by state and year after the 1994 Interstate Banking and Branching Efficiency Act.

Interpretation: The deregulation index is the core instrument. It changes banks' ability to diversify deposits and operate across state borders, thereby shifting the supply of bank credit.

2.3 House prices and housing supply

- County house prices come from repeat-sales house-price indexes.
- Housing stock measures come from county-level housing data.
- Housing supply elasticity comes from Saiz-style measures of geographic constraints.
- The paper distinguishes counties where construction is elastic from counties where construction is inelastic.

Interpretation: The same credit shock should affect prices more in places where new housing cannot be built easily, and quantities more in places where supply can expand.

2.4 Bank balance-sheet data

- Balance-sheet variables come from bank Call Reports and related regulatory sources.
- The paper studies mortgage rates, return on assets, cost of deposits, nonperforming loans, and deposit growth.
- These outcomes reveal whether deregulation relaxes bank funding constraints and improves credit terms.

Interpretation: If deregulation is a credit-supply shock, treated banks should collect more deposits, pay lower funding costs, and offer better mortgage terms.

3 Models and empirical specifications

3.1 Mortgage-credit first-stage regression

The baseline mortgage-credit specification is:

$$\ln L_{c,t} - \ln L_{c,t-1} = \beta_1 D_{s,t-1} + \beta_2 X_{c,t} + \alpha_c + \gamma_t + \varepsilon_{c,t}, \quad (1)$$

where $L_{c,t}$ is a mortgage outcome in county c , $D_{s,t-1}$ is the branching deregulation index for state s , $X_{c,t}$ are county controls, α_c are county fixed effects, and γ_t are year fixed effects.

Interpretation: The coefficient β_1 measures how mortgage credit growth changes after branching restrictions are relaxed.

3.2 Local projection impulse responses

For dynamic effects, the paper estimates:

$$\ln L_{c,t+i} - \ln L_{c,t+i-1} = \beta_1^{(i)} D_{s,t-1} + \beta_2 X_{c,t} + \alpha_c + \gamma_t + \varepsilon_{c,t}, \quad (2)$$

for horizons $i = 0, 1, \dots, 8$.

Interpretation: The local projection traces how mortgage credit responds over time. Credit growth rises on impact and then gradually fades.

3.3 House-price and housing-stock regressions

The house-price specification is:

$$\ln P_{c,t} - \ln P_{c,t-1} = \delta_1 D_{s,t-1} + \delta_2 D_{s,t-1} \times Elasticity_c + \delta_3 X_{c,t} + \alpha_c + \gamma_t + u_{c,t}. \quad (3)$$

The housing-stock specification replaces house-price growth with housing-stock growth and interacts deregulation with supply inelasticity.

Interpretation: Deregulation raises house prices more where supply is inelastic. In elastic markets, the response shifts toward construction instead of price appreciation.

3.4 Instrumental-variable house-price regression

The IV equation is:

$$\ln P_{c,t} - \ln P_{c,t-1} = \theta_1 (\ln L_{c,t} - \ln L_{c,t-1}) + \theta_2 X_{c,t} + \alpha_c + \gamma_t + \eta_{c,t}, \quad (4)$$

where mortgage-credit growth is instrumented with deregulation.

Interpretation: The IV estimates ask whether the part of mortgage-credit growth caused by deregulation raises house prices. This is the paper's central causal chain.

3.5 Bank balance-sheet regressions

The mechanism regressions use the same fixed-effect structure but replace the outcome with bank-level mortgage rates, return on assets, cost of deposits, nonperforming loans, or deposit growth.

Interpretation: These regressions test whether deregulation lowers bank funding costs and improves borrower credit terms.

4 Identification and empirical design

4.1 Why deregulation alone is not enough

- A simple comparison of deregulated and restricted states could be contaminated by state economic conditions.
- Deregulation could be endogenous to expected credit demand or housing demand.
- The paper therefore does not rely only on the timing of deregulation.

Interpretation: The empirical design is built around placebo groups that should respond to demand shocks but should not respond to a bank-branching credit-supply shock.

4.2 Placebo-lender design

- Commercial banks are affected by interstate branching rules.
- Independent mortgage companies, thrifts, and credit unions are legally unaffected.
- If deregulation captures mortgage demand, all lenders should respond similarly.
- If deregulation shifts bank credit supply, only commercial banks should respond.

Interpretation: Table 2 shows exactly this: commercial banks expand credit, while placebo lenders do not.

4.3 Bank-location design

- Out-of-state banks with local branches should benefit directly from branching deregulation.
- Out-of-state banks without branches in the deregulating state should not.
- In-state banks should not gain the same interstate geographic diversification benefit.

Interpretation: Table 3 shows that the effect is concentrated among out-of-state banks with local branches, tightening the credit-supply interpretation.

4.4 Housing-supply heterogeneity

- A credit shock should produce different outcomes depending on housing supply elasticity.
- In inelastic places, expanded mortgage demand translates into higher prices.
- In elastic places, the same shock translates more into new housing supply.

Interpretation: This heterogeneity links the credit channel to standard housing-market incidence logic.

4.5 What remains imperfect

- Deregulation is still a policy change, not a randomized experiment.
- Mortgage demand may respond to broad state-level conditions.
- HMDA data measure originated loans but not every margin of credit terms.
- The IV estimates depend on the exclusion restriction that deregulation affects house prices through bank credit supply.

Interpretation: The paper's strength is the cumulative evidence from placebo lenders, bank-location comparisons, supply-elasticity heterogeneity, and balance-sheet mechanisms.

5 Tables and figures

All visuals below are directly cropped from the paper and grouped compactly.

5.1 Table 1: Lender summary statistics

Read:

- Commercial banks account for a large share of mortgage applications and originations.
- Independent mortgage companies are also important, especially by 2005.

- Average loan sizes and applicants' incomes are broadly comparable across lender types, although IMCs tend to serve slightly smaller and lower-income borrowers.

Economic message: The placebo groups are economically meaningful lenders, not tiny irrelevant samples. This makes the placebo tests informative.

TABLE 1—LOANS BY COMMERCIAL BANKS, INDEPENDENT MORTGAGE COMPANIES, THRIFTS, AND CREDIT UNIONS

	Loans			
	Full sample 1994–2005	1995	2000	2005
<i>Number of applications received</i>				
Commercial banks	1,777	1,027	1,830	2,468
Independent mortgage companies	1,324	749	1,086	2,137
Thriffs and credit unions	738	534	740	921
<i>Number of loans originated</i>				
Commercial banks	1,437	865	1,361	2,083
Independent mortgage companies	973	526	734	1,659
Thriffs and credit unions	610	456	632	727
<i>Average loan originated (thousand of dollars)</i>				
Commercial banks	111	85	103	144
Independent mortgage companies	95	71	94	121
Thriffs and credit unions	114	83	115	143
<i>Average applicant's income (thousand of dollars)</i>				
Commercial banks	65	56	64	75
Independent mortgage companies	58	47	58	69
Thriffs and credit unions	67	54	70	75

Notes: Mean values of county-year pooled data. Conventional loans are for purchase of single-family owner occupied houses. Lenders are commercial banks, independent mortgage companies, or thriffs and credit unions. The sample includes US counties in urban areas for which mortgage data are available for the period 1994–2005.

5.2 Table 2 + Figure 1 + Figure 2: Commercial banks versus placebo lenders

Read:

- In Panel A of Table 2, deregulation increases commercial-bank mortgage originations by 0.028 and loan volume by 0.029.
- It also raises loan-to-income ratios by 0.029, but does not significantly change denial rates or the number of loans sold.
- In Panel B, placebo lenders have near-zero and insignificant responses.
- Figure 1 shows that the credit response is positive on impact and then fades over time.
- Figure 2 shows that deregulation can explain a large part of the rise in aggregate commercial-bank mortgage volume from 1994 to 2005.

Economic message: Deregulation shifts bank credit supply. The nonresponse of placebo lenders is the key evidence against a pure mortgage-demand explanation.

TABLE 2—COMMERCIAL BANKS VERSUS PLACEBO SAMPLES

	Number of originations	Volume	Number of denials	Loan to income ratio	Number sold
<i>Panel A. Commercial banks</i>					
Deregulation	0.028** (0.010)	0.029** (0.012)	0.013 (0.018)	0.029** (0.011)	0.008 (0.012)
Observations	10,992	10,992	10,923	10,922	10,689
Number of counties	1,018	1,018	1,018	1,018	1,018
Number of states	50	50	50	50	50
R ² within	0.122	0.132	0.389	0.124	0.112
<i>Panel B. Independent mortgage companies, thrifts, and credit unions</i>					
Deregulation	0.002 (0.008)	0.001 (0.008)	0.011 (0.015)	0.001 (0.008)	-0.015 (0.011)
Test	[0.029]	[0.038]	[0.721]	[0.042]	[0.268]
Observations	10,580	10,580	10,566	10,579	9,859
Number of counties	1,017	1,017	1,016	1,017	1,006
Number of states	50	50	50	50	50
R ² within	0.406	0.315	0.613	0.295	0.427

Notes: Dependent variables are the log change in the number or volume of mortgages originated, the number of applications denied, the loan to income ratio, and the number of mortgages originated and securitized. Regressors are: the Rice and Strahan (2010) Index of Interstate Branching Deregulation, a lagged dependent variable, current and lagged log change in county's income per capita, population, house price, and the Herfindahl index of loan concentration. All variables are defined in online Appendix Table OA1. The sample includes all US counties in urban areas for which mortgage data are available for the period 1994–2005. Panel A reports regression results for mortgage loans originated by commercial banks. Panel B reports regression results for the placebo sample of mortgage loans originated by independent mortgage companies, thrifts, or credit unions. The index of interstate branching deregulation ranges from 0 (most restricted) to 4 (least restricted). All regressions include county and year fixed effects. Standard errors are clustered by state. "Test" reports *p*-values associated with the null hypothesis that the coefficients in panel A are zero and equal to those in panel B.

*** Significant at the 1 percent level.

** Significant at the 5 percent level.

* Significant at the 10 percent level.

(a) Table 2: Commercial banks versus placebo lenders.

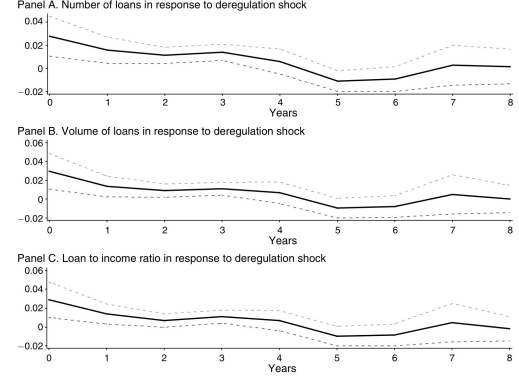


FIGURE 1. MORTGAGE CREDIT BY COMMERCIAL BANKS: IMPULSE RESPONSES TO BRANCHING DEREGULATION SHOCK

Note: Dashed lines are 90 percent confidence bands.

(b) Figure 1: credit impulse responses.

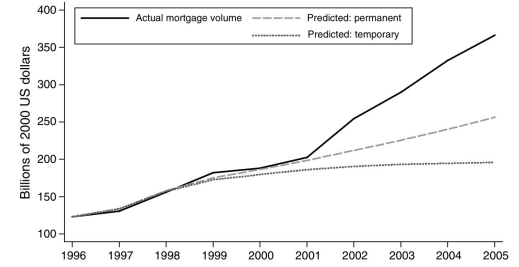


FIGURE 2. ACTUAL AND PREDICTED AGGREGATE MORTGAGE VOLUME BY COMMERCIAL BANKS

Note: This figure plots the aggregate real volume of mortgages for home purchase originated

(c) Figure 2: actual and predicted mortgage volume.

5.3 Table 3: Bank location and local branches

Read:

- Out-of-state banks with local branches respond strongly to deregulation.
- Their number of originations rises by 0.161, volume by 0.167, and loan-to-income ratio by 0.165.
- Out-of-state banks without local branches show much smaller responses.
- In-state banks do not expand credit in the same way.

Economic message: The credit expansion comes from the banks that actually gain from interstate branching. This supports the geographic-diversification/funding-cost mechanism.

5.4 Table 4: House prices, housing stock, and deregulation

Read:

- Deregulation raises house-price growth.
- The interaction with housing supply elasticity is negative: prices rise less where construction is easier.

TABLE 3—THE IMPORTANCE OF BANK LOCATION AND OF BANK BRANCHES

	Number of originations	Volume	Number of denials	Loan to income ratio	Number sold	Number local branches
<i>Panel A. Out-of-state banks: local branches</i>						
Deregulation	0.161** (0.060)	0.167*** (0.061)	0.069 (0.059)	0.165*** (0.061)	0.055 (0.082)	0.077** (0.030)
Observations	4,514	4,514	4,108	4,513	3,356	4,783
Number of counties	767	767	738	767	700	790
Number of states	49	49	49	49	47	49
R ² within	0.159	0.145	0.221	0.140	0.258	0.108
<i>Panel B. Out-of-state banks: no branches</i>						
Deregulation	0.016 (0.011)	0.015 (0.011)	0.012 (0.024)	0.014 (0.012)	0.010 (0.014)	—
Test	[0.012]	[0.017]	[0.429]	[0.018]	[0.673]	
Observations	9,988	9,988	9,926	9,987	9,771	—
Number of counties	1,018	1,018	1,018	1,018	1,017	—
Number of states	50	50	50	50	50	—
R ² within	0.160	0.122	0.475	0.112	0.188	—
<i>Panel C. In-state banks</i>						
Deregulation	0.003 (0.010)	−0.003 (0.011)	0.005 (0.015)	−0.003 (0.011)	−0.025 (0.019)	0.001 (0.007)
Test	[0.016]	[0.024]	[0.431]	[0.024]	[0.400]	[0.032]
Observations	9,841	9,841	9,339	9,840	9,175	9,019
Number of counties	1,017	1,017	1,005	1,017	1,006	978
Number of states	50	50	50	50	50	50
R ² within	0.025	0.391	0.067	0.042	0.097	0.047

Notes: Dependent variables are the log change in the number or volume of mortgages originated, number of applications denied, loan to income ratio, number of mortgages originated and securitized, and the log change in the number of local branches (in panels A and C). Regressors are the Rice and Strahan (2010) Index of Interstate Branching Deregulation, a lagged dependent variable, current and lagged log change in county's income per capita, population, house price, and the Herfindahl index of loan concentration. All variables are defined in online Appendix Table OA1. The sample includes all US counties in urban areas for which mortgage data are available for the period 1994–2005. Panel A reports results for out-of-state-banks with local branches, i.e., lending in a location outside of the state where they are headquartered, and where they have at least one branch. Panel B reports results for out-of-state-banks without local branches, i.e., lending in a location outside of the state where they are headquartered, and where they have no branch. Panel C reports regression results for in-state banks, i.e., lending for a property in the state they are headquartered. The index of interstate branching

TABLE 4—HOUSE PRICES, HOUSING STOCK, AND DEREGULATION

	House prices			Housing stock	
	(1)	(2)	(3)	(4)	(5)
Index of interstate branching deregulation	0.0103** (0.004)	0.0242*** (0.005)	0.0122*** (0.002)	0.00176*** (0.0005)	0.00028 (0.0002)
Index of interstate branching deregulation × house supply elasticity		−0.008*** (0.002)	−0.005*** (0.000)		
Index of interstate branching deregulation × house supply inelasticity				−0.00258*** (0.0006)	−0.00083*** (0.0003)
Lagged house price			0.482*** (0.029)		
Lagged housing stock					0.6204*** (0.062)
Income per capita			0.043 (0.044)		0.110*** (0.026)
Lagged income per capita			0.070*** (0.026)		0.0279*** (0.009)
Population			0.590*** (0.124)		0.777*** (0.029)
Lagged population			0.243* (0.129)		−0.4962*** (0.069)
Herfindahl			0.001 (0.002)		0.0006* (0.0003)
Lagged Herfindahl			0.002 (0.001)		0.0004* (0.0002)
Observations	12,214	10,627	9,735	10,627	9,735
Number of counties	1,021	887	886	887	886
Number of states	50	48	48	48	48
R ²	0.199	0.202	0.428	0.048	0.737

Notes: County level linear regressions of the log change in house prices (columns 1 to 3) or the log change in the housing stock (columns 4 and 5) on the Rice and Strahan (2010) Index of Branching Deregulation and its interaction with the Saiz's Index of Housing Supply Elasticity (columns 1 to 3) or its inverse (columns 4 and 5). All variables are defined in online Appendix Table OA1. The sample includes all US urban counties for which mortgage data are available for the period 1994–2005. The index of interstate branching deregulation ranges from 0 (most restricted) to 4

- Deregulation raises housing-stock growth, but less in supply-inelastic areas.
- The results survive controls for lagged house prices, housing stock, income, population, and concentration.

Economic message: Credit supply increases housing demand. Whether the adjustment shows up as prices or quantities depends on local housing supply conditions.

5.5 Table 5 + Figures 3 and 4: IV credit channel and house prices

Read:

- Instrumented number of loans has a house-price coefficient of 0.141.
- Instrumented loan volume has a coefficient of 0.134.
- Instrumented loan-to-income ratio has a coefficient of 0.120.
- First-stage F-statistics are around 9–10, and underidentification p-values are zero.
- Figure 3 shows positive house-price responses to instrumented credit shocks.
- Figure 4 shows that deregulation-induced credit can explain a large share of the actual rise in real house prices.

Economic message: The IV evidence closes the causal chain: deregulation raises credit supply, and the induced credit expansion raises house prices.

TABLE 5—INSTRUMENTAL VARIABLE REGRESSIONS FOR HOUSE PRICES

	House prices		
	(1)	(2)	(3)
Instrumented number of loans	0.141** (0.069)		
Instrumented loan volume		0.134** (0.062)	
Instrumented loan to income ratio			0.120** (0.057)
Lagged house price	0.507*** (0.029)	0.457*** (0.039)	0.473*** (0.037)
Income per capita	0.021 (0.031)	0.007 (0.029)	0.061*** (0.020)
Lagged income per capita	0.076** (0.037)	0.056 (0.037)	0.116*** (0.042)
Population	-0.071 (0.334)	-0.067 (0.310)	0.277 (0.243)
Lagged population	0.220* (0.131)	0.195 (0.142)	0.094 (0.299)
Herfindahl	0.021** (0.010)	0.018** (0.009)	0.015* (0.008)
Lagged Herfindahl	0.004 (0.004)	0.003 (0.004)	0.004 (0.004)
First stage <i>F</i> -test of excluded instruments	10.13	10.31	8.59
Underidentification test (<i>p</i> -values)	0.000	0.000	0.000
Observations	11,107	11,107	11,107
Number of counties	1,018	1,018	1,018
Number of states	50	50	50

Notes: Second stage county level linear regressions of an IV specification of the log change in house prices on the number of loans, the loan volume, or the loan to income ratio of commercial banks. Number of loans, loan volume, and loan to income ratio are instrumented with the Rice and Strahan (2010) Index of Branching Deregulation. All variables are defined in online Appendix Table OA1. The sample includes all US urban counties for which mortgage and house price data is available for the period 1994–2005. Regressions include county and year fixed effects. Standard errors are robust to heteroskedasticity and autocorrelation.

(a) Table 5: IV regressions for house prices.

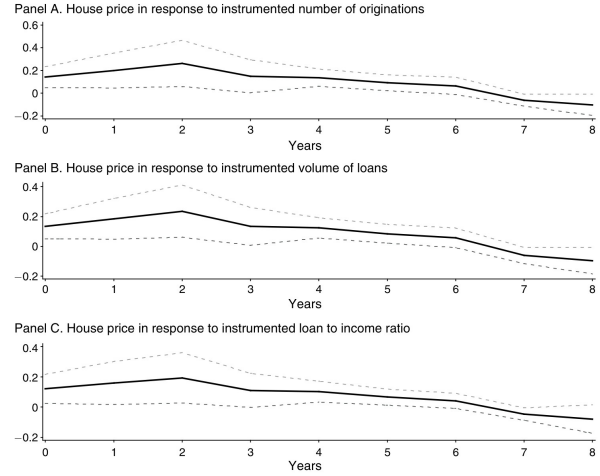


FIGURE 3. HOUSE PRICES: IMPULSE RESPONSES TO INSTRUMENTED CREDIT SHOCK

Note: Dashed lines are 90 percent confidence bands.

(b) Figure 3: house-price impulse responses.

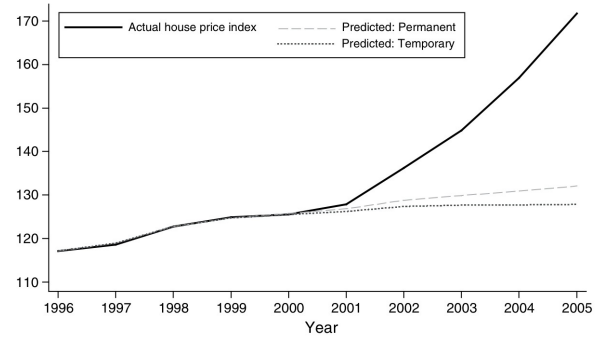


FIGURE 4. ACTUAL AND PREDICTED REAL HOUSE PRICE INDEX

Note: This figure plots the aggregate real house price index between 1994 and 2005, and the two predicted indexes (dashed and dotted lines) as described in Section IIIB.

(c) Figure 4: actual and predicted real house-price index.

5.6 Table 6: Bank balance-sheet mechanism

Read:

- Deregulation lowers mortgage rates by about 0.00483 in the full sample and 0.00336 in the border sample.
- It lowers the cost of deposits and raises deposit growth.
- It has no meaningful effect on return on assets or nonperforming loans.

Economic message: The mechanism is funding-side credit supply: deregulated banks get cheaper deposits, pass savings to borrowers, and expand credit without obvious deterioration in loan quality.

TABLE 6—COMMERCIAL BANKS: BALANCE SHEET DATA

	Mortgage rate	ROA	Cost of deposits	NPL	Deposit growth
<i>Panel A. Full sample</i>					
Deregulation	−0.00483*** (0.00123)	0.00003 (0.00005)	−0.00016*** (0.00006)	0.00013 (0.00016)	0.0436** (0.0167)
Observations	11,136	11,147	11,147	10,607	10,020
Number of counties	1,018	1,018	1,018	1,018	1,018
Number of states	50	50	50	50	50
R ² within	0.531	0.357	0.955	0.070	0.274
<i>Panel B. Border sample</i>					
Deregulation	−0.00336*** (0.00078)	0.00001 (0.00004)	−0.00023*** (0.00008)	0.00002 (0.00027)	0.0533*** (0.01736)
Observations	2,921	2,924	2,924	2,837	2,631
Number of counties	267	267	267	267	267
Number of states	35	35	35	35	35
R ² within	0.543	0.382	0.946	0.122	0.299

Notes: Dependent variables are mortgage rate (the ratio of interest and fee income on mortgage loans to total mortgage loans), ROA (net income over assets), cost of deposits (interest expenses on deposit to total deposits), NPL (nonperforming mortgage loans to total mortgage loans), deposit growth (log change in total deposits). The regressors are: the Rice and Strahan (2010) Index of Interstate Branching Deregulation, bank equity capital to assets, liquid assets to total assets, total loans to assets, as well as current and lagged log change in county's income per capita, population, house price, and the Herfindahl index of loan concentration. All variables are defined in online Appendix Table OA1. Panel A reports regressions for the full sample of US counties in urban areas. Panel B reports regression results for US counties in MSAs straddling two or more US states. The index of interstate branching deregulation ranges from 0 (most restricted) to 4 (least restricted). All regressions include county and year fixed effects. Standard errors are clustered by state.

***Significant at the 1 percent level.

**Significant at the 5 percent level.

*Significant at the 10 percent level.

the Rice and Strahan (2010) branching deregulation index over three-year intervals (Figure B1), and the geographic coverage of the urban counties used in the main regression analysis (Figure B2).

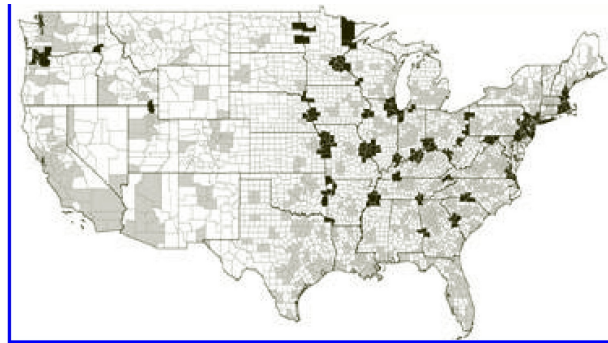
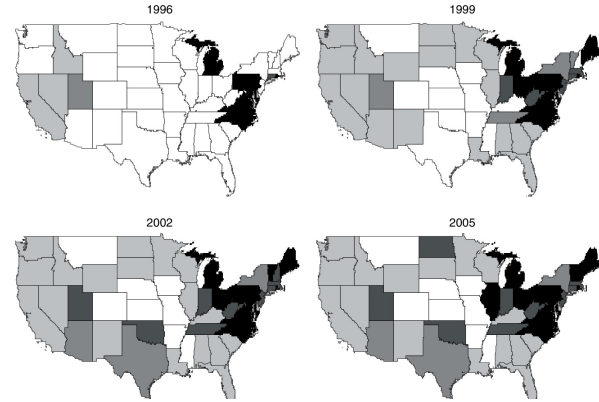


FIGURE B2. SAMPLE OF US URBAN COUNTIES (Gray)
AND COUNTIES IN MSA BORDERING TWO OR MORE STATES (Black)

Source: HMDA and Moody's.com

Read:

- The deregulation index varies across states and over time.
- Relaxation is not uniform geographically, which creates the policy variation used in the paper.

Economic message: The maps show why a county and year fixed-effect design can exploit staggered regulatory changes.

6 Short summary

- The paper studies whether mortgage credit supply affects house prices.
- The instrument is post-1994 interstate bank-branching deregulation.
- The paper does not rely on deregulation being randomly timed; it uses placebo lenders and bank-location comparisons.
- Commercial banks expand mortgage credit after deregulation.

- Independent mortgage companies, thrifts, and credit unions do not respond in the same way.
- Out-of-state banks with local branches respond most strongly.
- Deregulation raises house prices, especially in supply-inelastic counties.
- In supply-elastic counties, deregulation increases the housing stock instead.
- IV estimates show that deregulation-induced credit growth raises house prices.
- Balance-sheet evidence shows lower mortgage rates, lower deposit costs, and higher deposit growth.
- The main lesson is that bank credit supply can move housing asset prices.

7 Conclusion

7.1 Core conclusion

Favara and Imbs show that an exogenous expansion in mortgage credit supply raises house prices. They identify credit supply with interstate branching deregulation, validated through placebo lenders and location-based bank comparisons.

7.2 Economic intuition

Branching deregulation allows banks to diversify deposit funding across states and reduce funding costs. Lower funding costs relax banks' ability to originate mortgages. Expanded mortgage credit increases housing demand. In places where housing supply cannot expand easily, the demand shock becomes a price boom. In places where supply is elastic, the housing stock expands instead.

7.3 Takeaway

The paper provides a clean empirical link from bank regulation to credit supply to housing prices. It shows that housing booms can originate from changes in financial intermediation, not just from borrower demand or expectations.